

NAIROBI DAGORETTI, Kenya

WMO: 637410

Lat: 1.3013S

Lon: 36.7597E

Elev: 5900

StdP: 11.82

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.0	52.1	38.6	42.8	76.6	41.3	47.7	74.2	11.6	68.5	10.0	67.9	0.1	160	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.0	81.8	59.1	80.4	59.1	78.9	59.6	65.0	72.5	64.3	72.0	63.7	71.2	9.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	107.4	66.6	62.3	105.0	66.0	61.5	101.9	65.2	33.5	72.7	32.8	71.8	32.3	71.1	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.6	11.5	46.3	85.1	2.0	2.4	44.9	86.8	43.8	88.3	42.6	89.6	41.2	91.4		
(5)			44.8	69.8	2.2	2.5	43.2	71.6	41.9	73.1	40.6	74.5	39.0	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	67.0	68.3	69.0	68.1	65.7	63.5	62.1	62.8	65.5	67.0	66.3	66.2
	DBStd	2.99	1.90	1.90	2.20	1.99	1.82	2.18	2.43	2.30	2.66	2.17	2.11	1.69
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	296	5	1	1	14	54	94	75	25	8	11	7	
	CDD50	5820	528	512	589	544	486	406	376	396	464	528	488	503
	CDD65	639	68	93	125	95	35	10	4	5	39	71	49	45
	CDH74	2754	442	653	687	228	34	28	33	48	213	247	72	69
	CDH80	145	13	48	58	12	0	0	0	0	7	6	1	0
Wind	WSAvg	3.9	5.2	4.9	5.6	3.8	2.4	1.9	1.8	2.2	3.5	4.9	4.9	5.3
Precipitation	PrecAvg	41.1	2.1	1.9	3.7	9.6	7.4	1.5	0.7	0.9	1.2	2.3	5.8	3.9
	PrecMax	65.0	7.8	7.2	11.8	27.8	20.9	5.6	2.7	3.0	5.4	7.7	24.5	14.9
	PrecMin	19.0	0.0	0.0	0.2	0.8	0.1	0.2	0.0	0.1	0.0	0.1	0.8	0.3
	PrecStd	11.1	2.2	2.0	2.8	6.0	4.7	1.3	0.6	0.8	1.1	1.9	4.3	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.3	82.8	83.5	81.5	76.9	77.1	77.1	77.4	80.7	80.7	78.6	77.3
		MCWB	58.0	58.2	58.6	60.6	62.5	61.6	60.4	59.9	59.0	59.6	60.5	61.2
	2%	DB	80.2	81.3	81.9	78.4	75.0	73.8	74.5	75.4	78.5	78.7	75.9	75.7
		MCWB	58.3	58.5	59.4	61.4	62.6	61.7	60.1	59.8	59.4	59.5	61.3	61.1
	5%	DB	78.5	80.3	80.0	76.7	73.3	71.9	72.2	73.3	76.6	76.9	74.4	74.4
		MCWB	59.0	58.6	59.7	62.0	62.1	61.0	59.4	59.1	59.3	59.7	61.4	61.2
	10%	DB	76.9	78.5	78.4	74.9	71.7	70.0	70.0	71.2	74.6	75.0	72.8	73.1
		MCWB	59.4	59.1	60.1	62.2	61.7	60.4	59.1	58.9	59.1	59.8	61.4	61.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	65.7	65.1	66.0	65.5	64.7	62.9	63.0	63.5	63.8	64.7	64.8
		MCDB	74.0	73.6	74.7	72.8	72.2	71.4	72.0	71.7	73.4	71.7	72.1	71.2
	2%	WB	62.9	64.3	63.8	64.8	64.3	63.1	61.6	61.7	62.0	62.8	63.5	63.6
		MCDB	72.4	73.0	73.1	71.8	71.1	70.0	70.6	71.1	72.2	70.8	70.7	70.2
	5%	WB	62.1	63.0	62.9	64.1	63.4	62.2	60.5	60.3	61.0	62.0	62.8	62.9
		MCDB	71.5	71.4	71.9	71.1	69.7	68.8	69.3	69.7	71.6	69.8	69.6	69.6
	10%	WB	61.3	62.2	62.3	63.4	62.7	61.5	59.6	59.5	60.1	61.4	62.1	62.3
		MCDB	70.8	70.6	71.0	70.2	68.5	67.9	67.9	68.2	70.4	68.9	68.6	68.8
Mean Daily Temperature Range	5% DB	MDBR	20.9	21.6	19.0	15.5	13.4	14.6	16.1	16.6	19.1	16.8	14.3	14.8
		MCDBR	23.6	24.9	22.1	17.7	15.8	17.6	20.8	21.9	22.3	20.1	16.4	16.8
		MCWBR	7.3	7.4	5.9	5.2	5.1	7.3	8.7	8.4	7.1	5.4	4.8	5.3
	5% WB	MCDBR	18.9	18.6	17.6	15.3	13.5	15.8	18.1	18.6	19.6	15.4	14.3	14.2
		MCWBR	6.8	6.9	5.7	5.3	5.3	7.1	8.2	8.0	7.3	5.2	5.2	5.2
Clear-Sky Solar Irradiance	taub	0.362	0.390	0.373	0.321	0.320	0.356	0.370	0.357	0.338	0.350	0.344	0.355	
	taud	2.453	2.354	2.432	2.646	2.640	2.483	2.403	2.434	2.495	2.464	2.520	2.492	
	Ebn at Noon	307	299	301	310	303	287	285	295	307	307	311	308	
	Edh at Noon	37	42	38	30	29	33	36	37	36	37	35	36	
All-Sky Solar Radiation	RadAvg	2055	2150	2086	1840	1686	1501	1487	1581	1894	1924	1785	1904	
	RadStd	135	110	137	100	63	102	138	113	104	96	97	118	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page